

ROHIT MAHALI

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Education

B.Tech + M.Tech (Integrated), Computer Science & Engineering

2022 – 2027

Central University of Jharkhand

Spec: Machine Learning & Data Science

Technical Skills

Data Analysis & Databases: Advanced SQL (Joins, Window Functions, CTEs, Query Optimization), Relational DBs, Normalization

Programming: Python (Pandas, NumPy, Matplotlib, Scikit-learn, PyMuPDF), C

Visualization & BI: Power BI, Streamlit (Dashboarding), Matplotlib, Seaborn

Data Engineering: ETL Pipelines, Data Cleaning, Regex (Pattern Matching), Unstructured Data Parsing

Tools & Cloud: Git/GitHub, Google Cloud Platform (GCE), Excel (Pivot Tables, VLOOKUP)

Experience

Indigenous Language Development Intern

Remote/Hyderabad

IndicWiki Project, IIIT Hyderabad

- Utilized **complex querying techniques** (Wiki Query) to extract, filter, and structure data from large-scale repositories, ensuring high data integrity for downstream tasks.
- Managed relational data mapping to ensure consistency across Wikipedia sister projects, identifying and resolving schema discrepancies.
- Performed rigorous data validation and quality checks on datasets to ensure accuracy for low-resource languages.

Projects & Research

Dia-RAG: Dialect-Aware Retrieval Architecture
(Ongoing Research)

<https://github.com/rohitmahali01/Dia-Rag>

- Research Proposal:** Authored a comprehensive research paper proposing "Dia-RAG," a novel framework to solve the "**Standardization Bottleneck**" and lexical erasure in low-resource dialectal RAG systems[cite: 81, 82].
- System Design:** Designed a theoretical architecture featuring a "**Diversity-Sensitive Decoupler**" to split queries into invariant/variant components and a **Dictionary-Augmented Retrieval** layer to preserve cultural nuance[cite: 181, 186, 197].
- Experimental Design:** Formulated the "**Santali-CultureQA**" evaluation benchmark and proposed the **Cultural Specificity Score (CSS)** metric to quantitatively measure hallucination in dialectal generation[cite: 223, 256].

Santali Language Suite (Data Cleaning + ASR)

<https://github.com/rohitmahali01/Santali-Tokenizer> |
<https://github.com/rohitmahali01/Whisper-Santali-Asr>

- Orchestrated the end-to-end data pipeline for a low-resource language model, processing large-scale audio datasets from the Mozilla Common Voice repository.
- Implemented rigorous **data cleaning** and normalization workflows using Python to handle noise, silence removal, and text-audio alignment, significantly improving dataset quality.
- Fine-tuned the Whisper ASR model and evaluated performance using **Word Error Rate (WER)**.

Smart Tax Platform (AI Document Analysis)

<https://github.com/rohitmahali01/Smart-Tax-Platform-with-AI-Document-Analysis>

- Engineered a financial data extraction engine using **Python (PyMuPDF)** to parse unstructured PDF salary slips; implemented robust logic to handle **null values** and missing fields automatically.
- Designed intelligent **Regex** pattern matching algorithms to identify and extract key financial indicators (Gross Salary, HRA, TDS) across varying document layouts with high precision.
- Integrated **Google Generative AI** to analyze extracted data and generate comparative reports on tax regimes.

Hackathons

LIT-RAG — HackRx 6.0 Hackathon Semifinalist (Top 38/9000)

- Led a team of 5 to design a high-speed **ETL pipeline** for processing unstructured document data, reducing data retrieval time to < 15s.
- Orchestrated the extraction, transformation (embeddings), and loading of data into Pinecone Vector DB, utilizing Python multiprocessing for concurrency.

Ol Chiki Transliterator (Interactive Dashboard)

<https://github.com/rohitmahali01/Olchiki-Transliterator>

- Designed and deployed an interactive **Streamlit dashboard** to visualize real-time text transliteration between Ol Chiki, Devanagari, and Latin scripts.